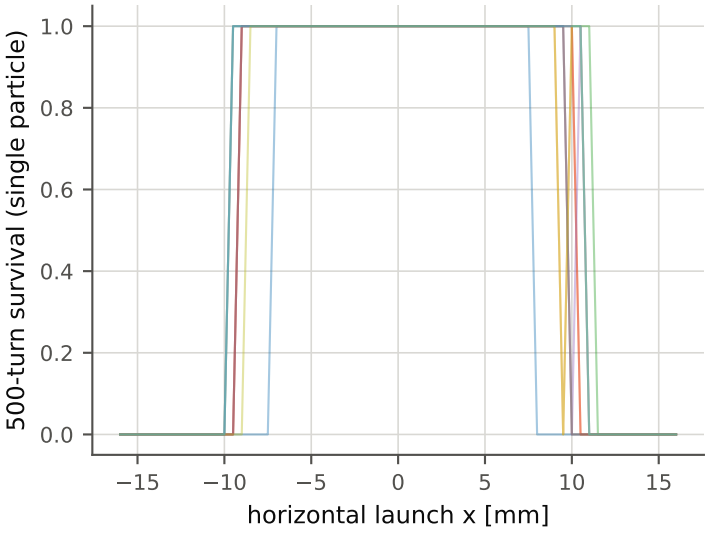
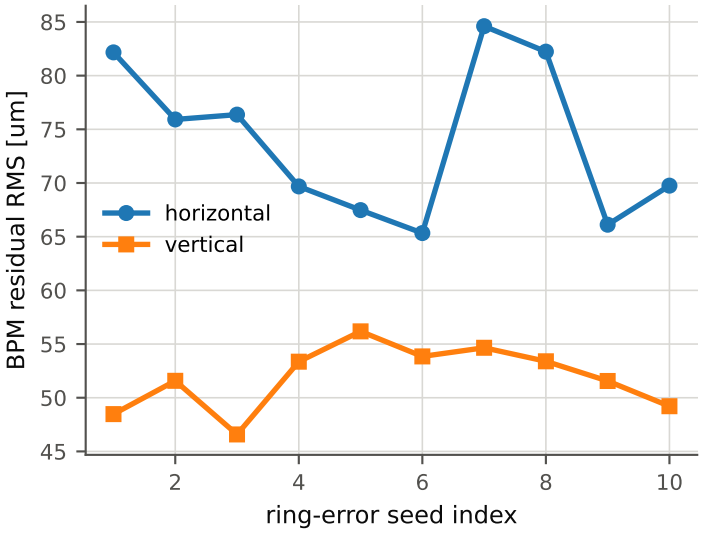

500-turn survival (single particle)
1.0
0.8
0.6
0.4
0.2
0.0
−15
−10
−5
0
5
10
15
horizontal launch x [mm]


BPM residual RMS [um]
85
80
75
70
65
60
55
50
45
horizontal
vertical
2
4
6
8
10
ring-error seed index